

PRANAV PARASAR

+91 72770 07271 | pranavparasar2@gmail.com
https://linkedin.com/in/pranav-parasar-28a758328 | https://github.com/pranavparasar
Bhubaneswar, Odisha, India

PROFESSIONAL SUMMARY

Electronics and Communication Engineering student at ITER, Siksha 'O' Anusandhan with a strong foundation in Java, DSA, SQL, HTML/CSS, Automation, AWS, Embedded Systems, FPGA Design, VLSI, Signal Processing, and Digital Electronics. Experienced in designing hardware-software integrated systems using MATLAB, Simulink, VHDL. Hands-on experience through academic projects and industrial training in automation, Digital Twin technology, Industry 5.0, IoT-enabled systems, and embedded system development. Seeking opportunities to contribute technical expertise and problem-solving skills in embedded systems, semiconductor, and automation industries.

EDUCATION

Bachelor of Technology (B. Tech) – Electronics and Communication Engineering	2023 – Present
<i>Institute of Technical Education and Research (ITER), S 'O' A University</i>	Bhubaneswar, Odisha
Senior Secondary Education (CBSE – PCM)	2023
<i>Star International School</i>	Ranchi, Jharkhand
Secondary Education (CBSE)	2021
<i>Jawahar Vidya Mandir, Shyamali</i>	Ranchi, Jharkhand

TECHNICAL SKILLS

Programming Languages: Java, Data Structures & Algorithms, Python, C++, SQL, VHDL& HTML/CSS
Embedded & Electronics: Embedded Systems, FPGA Programming, Digital Electronics, Analog Electronics, VLSI Design, Signal Processing, Digital Twin Systems, Automation& IoT
Software & Engineering Tools: MATLAB, Simulink, Multisim, Microsoft Excel, Microsoft PowerPoint& Google Sheets
Web Technologies & Database: AWS, Node.js, MongoDB, Frontend& Backend Development

ACADEMIC PROJECTS

- ECG Signal Processing and QRS Detection in MATLAB (2026)**
 - Designed an ECG preprocessing pipeline for noise removal and signal enhancement.
 - Implemented Pan-Tompkins algorithm for accurate QRS complex and R-peak detection.
- DTMF Tone Detection using MATLAB (2026)**
 - Developed a Dual-Tone Multi-Frequency (DTMF) decoder for audio signals.
 - Applied FFT to identify frequency components and enable accurate real-time keypad tone detection.
- Analog Computer Circuit for Solving Mathematical Equations using IC-555 (2025)**
 - Designed analog computing circuits to solve differential equations.
 - Implemented operational amplifier-based integrator and summing circuits; verified through waveform analysis.
- FPGA-Based Seven Segment Display Controller in VHDL (2024)**
 - Designed and implemented a multiplexed seven-segment display controller on the BASYS-3 FPGA board.
 - Optimized timing constraints for stable and efficient multi-digit display operation.
- Tea Vending Machine using IC555 (2023)**
 - Designed an automated tea dispensing system using NE555 timer and relay-operated pumps.
 - Controlled ingredient dispensing based on user input; developed a low-cost automation system without microcontroller.

INTERNSHIP EXPERIENCE

Steel Authority of India Limited (SAIL), CET, PC&A	June 2026
Project Title: Digital Twin Integration for Steel Plant Automation	Ranchi, Jharkhand
- Developed a Digital Twin-based industrial automation framework integrating PLC, SCADA, IoT-enabled data acquisition, and cloud analytics for real-time process monitoring. - Applied Industry 5.0 concepts through intelligent automation, real-time data analytics, and human-centric decision support to enhance plant efficiency and operational reliability. - Contributed to IoT-driven predictive maintenance and process optimization, supporting improved equipment reliability, plant safety, and overall productivity.	

CERTIFICATIONS & ACHIEVEMENTS

- Infosys Springboard – CSS3 Fundamentals
- Infosys Springboard – HTML5: The Language
- Embedded System Designing – Altium
- Microsoft Learn – ASP.NET Core Fundamentals

LEADERSHIP & EXTRACURRICULAR ACTIVITIES

- Active member of JAAGO Club, contributing to educational initiatives for underprivileged children.
- Active member of S 'O' A Flying Community, assisting in event planning and technical activities.

REFERENCES

J.K. REDDY
Title: Deputy General Manager (PC&A)
Company: Steel Authority of India Limited (SAIL)
Location: Ranchi, JH | Phone: +918986880529 | Email: kj.reddy@sail.in
Relationship: Guided me in my internship project on Digital Twin Integration for Steel Plant Automation.